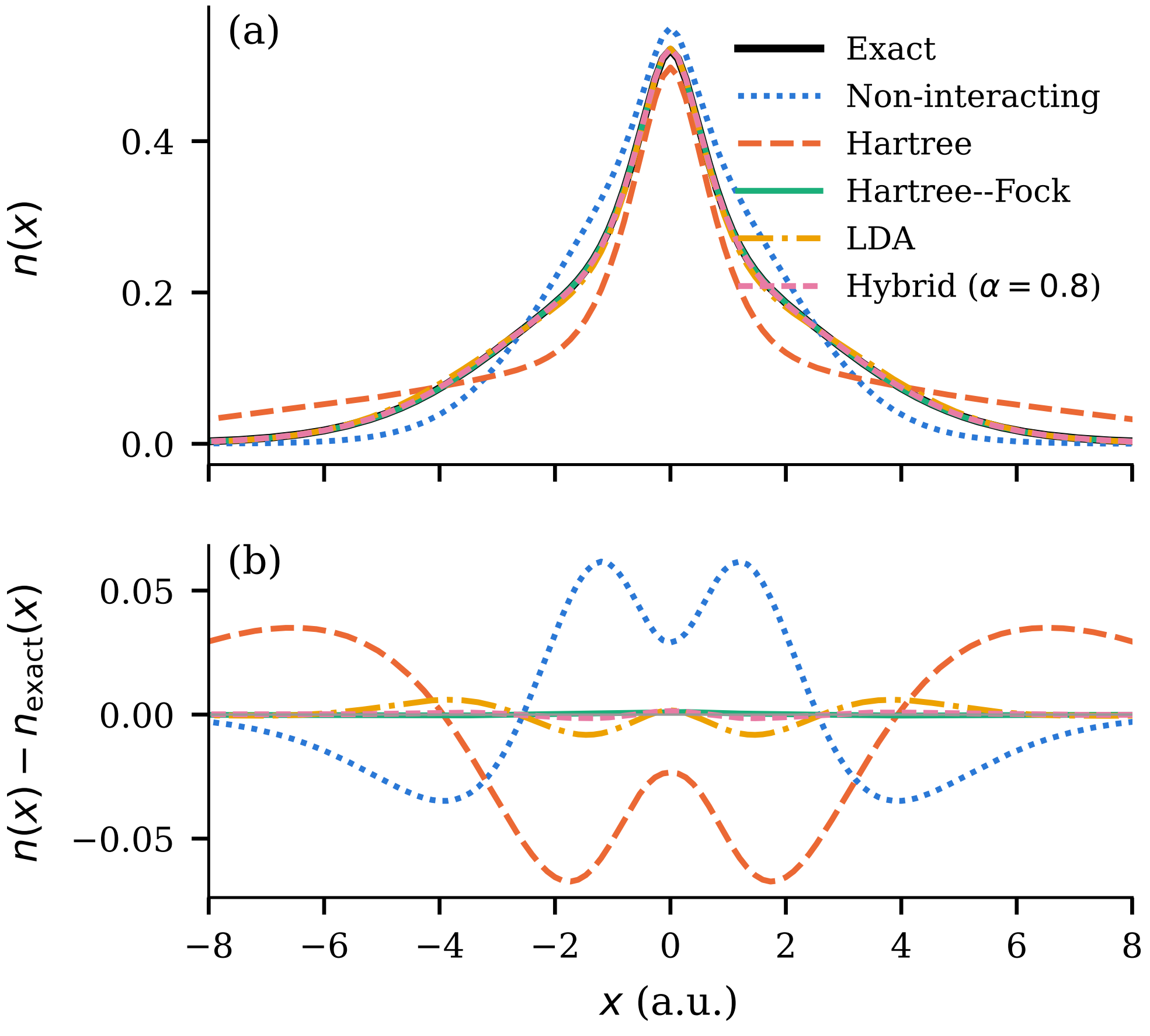
(a)
Exact
Non-interacting
Hartree
Hartree--Fock
LDA
Hybrid ($\alpha = 0.8$)
$n(x)$
0.0
0.2
0.4
(b)
$n(x) - n_{\mathrm{exact}}(x)$
0.05
0.00
−0.05
−8
−6
−4
−2
0
2
4
6
8
x (a.u.)